

Task 1:

Forked GitHub Repo

The screenshot shows a GitHub repository page for 'CS487-PA4' by user 'MmaazBarki'. The repository is a fork of 'karmamg/CS487-PA4'. The main branch is 'main' with 1 branch and 0 tags. The repository is 8 commits ahead of the upstream. The commit history shows a series of commits, including 'Refactor Azure deployment workflow for Node.js 20', 'Added: Starter Pack for PA4', 'Fix: Migrate Storage auth to Managed Identity to bypass Shar...', and 'Added: Starter Pack for PA4'. The repository contains files like '.github/workflows', 'docs', 'function-app', 'report-job', 'validate-api', 'webapp', '.gitignore', 'README.md', and 'SUBMISSION_TEMPLATE.md'. The right sidebar shows the 'About' section, 'Releases', 'Packages', 'Contributors', and 'Languages'.

TaskFlow Cloud Pipeline

Web App Overview: Running Status

The screenshot shows the 'Overview' page for a Web App named 'pa4-26100414'. The status is 'Running'. The page displays various details about the web app, including its resource group, location, subscription, and tags. The 'Essentials' section shows the resource group as 'rg-sp26-26100414', the location as 'Sweden Central', the subscription as 'Microsoft Azure - Ali Khawaja', and the subscription ID as '67e93b84-fe08-452c-80ea-175d0a3eca56'. The 'Tags' section shows 'Add tags'. The 'Default domain' is 'pa4-26100414-gnc0b8ceatd9acc5swedencentral-01.azurewebsites.net'. The 'App Service Plan' is 'pa4-26100414 (F1: 1)'. The 'Operating System' is 'Linux'. The 'Health Check' is 'Not Configured'. The 'GitHub Project' is 'https://github.com/MmaazBarki/CS487-PA4'.

Deployment Center: GitHub Source

Search

Settings

Containers (new)

Logs

FTPS Credentials

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Microsoft Defender for Cloud

Events (preview)

Log stream

Resource visualizer

Deployment

Deployment slots

Deployment Center

Settings

Environment variables

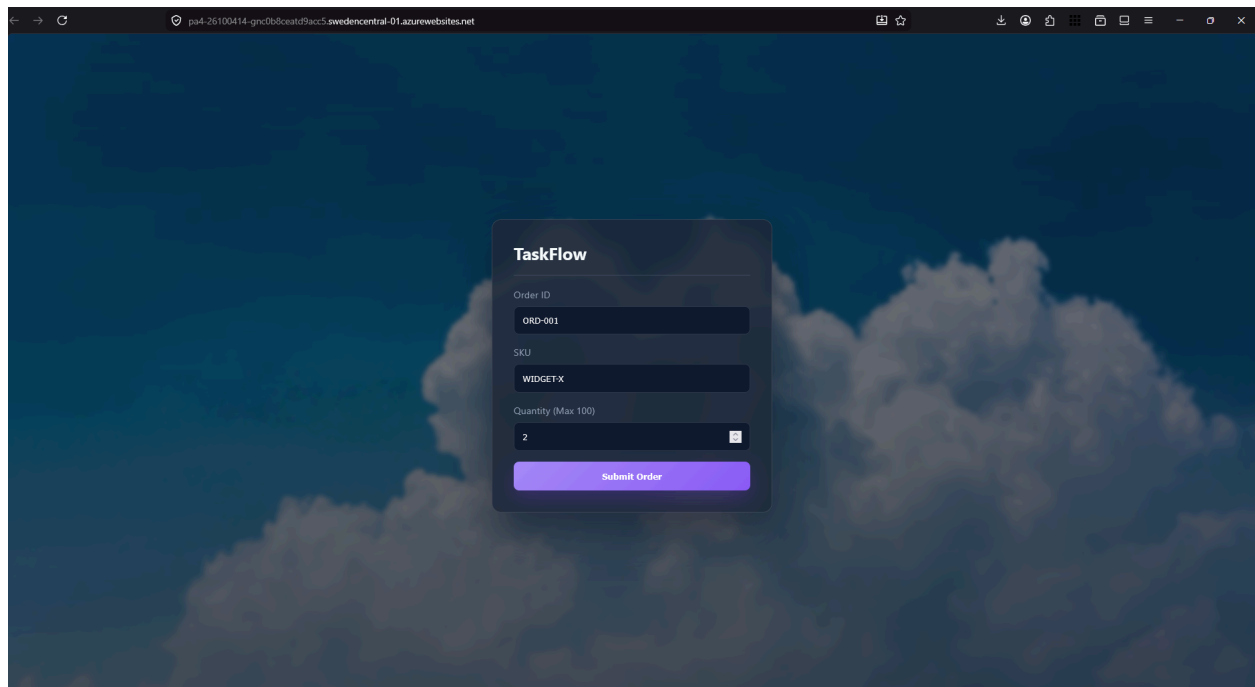
Configuration

Save Discard Refresh Browse Sync Send us your feedback

Deploy and build code from your preferred source and build provider. [Learn more](#)

Source	GitHub Disconnect
GitHub	
Organization	MmaazBarki
Repository	CS487-PA4
Branch	main ↗
Build	
Build provider	GitHub Actions
Runtime stack	Node
Version	22-lts

Dashboard Loading in Browser:



Configured Application Settings:

Home > pa4-26100414

 **pa4-26100414** | Environment variables ☆ ...
Web App

Search

App settings Connection strings

Overview Activity log Access control (IAM) Tags Diagnose and solve problems Microsoft Defender for Cloud Events (preview) Log stream

Search + Add Refresh Show values Advanced edit Pull reference values

Name	Value
FUNCTION_START_URL	Show value
FUNCTION_STATUS_URL	Show value

Task 2:

ACR Overview Success:

pa426100414 Container registry Check AKS clusters pulling from this registry

Search Move Delete

Overview

Activity log Access control (IAM) Tags Quick start Resource visualizer

Essentials

Resource group (move) : rg-sp26-26100414
Location : Sweden Central
Subscription (move) : Microsoft Azure - Ali Khawaja
Subscription ID : 67e93b84-fe08-452c-80ea-175d0a3eca56
Soft delete (Preview) : Disabled
Tags (edit) : Add tags

Login server : pa426100414.azurecr.io
Creation date : 5/5/2026, 11:47 PM GMT+5
Provisioning state : Succeeded
Pricing plan : Basic
Domain name label scope : Unsecure

Successful Docker Builds

```
(base) PS C:\Users\maazb\OneDrive\Desktop\Cloud-Dev\Assignments\Assignment4\CS487-PA4> docker build -t validate-api:latest ./validate-api
[+] Building 65.6s (11/11) FINISHED
docker:desktop-linux

(base) PS C:\Users\maazb\OneDrive\Desktop\Cloud-Dev\Assignments\Assignment4\CS487-PA4> docker build -t report-job:latest ./report-job
[+] Building 17.7s (10/10) FINISHED
docker:desktop-linux

(base) PS C:\Users\maazb\OneDrive\Desktop\Cloud-Dev\Assignments\Assignment4\CS487-PA4> docker build -t func-app:latest ./function-app
[+] Building 2399.9s (9/9) FINISHED
docker:desktop-linux
```

Successful local validator test

```
(base) PS C:\Users\maazb\OneDrive\Desktop\Cloud-Dev\Assignments\Assignment4\CS487-PA4> curl.exe -X POST http://localhost:8080/validate -H "Content-Type: application/json" -d '{"order_id":"","items":[{"sku":"","quantity":1}]}
{"valid":true,"reason":"ok","order_id":"LOCAL-1"}
```

Image Tagging Commands:

docker tag validate-api:latest pa426100414.azurecr.io/validate-api:v1

docker tag report-job:latest pa426100414.azurecr.io/report-job:v1

docker tag func-app:latest pa426100414.azurecr.io/func-app:v1

Image Pushing Commands:

docker push pa426100414.azurecr.io/validate-api:v1

docker push pa426100414.azurecr.io/report-job:v1

docker push pa426100414.azurecr.io/func-app:v1

Successful Image Push:

```
(base) PS C:\Users\maazb\OneDrive\Desktop\Cloud-Dev\Assignments\Assignment4\CS487-PA4> docker push pa426100414.azurecr.io/validate-api:v1
The push refers to repository [pa426100414.azurecr.io/validate-api]
a17f4a15dfff7: Pushed
7ccd73948dde: Pushed
91ff8760033c: Pushed
3531af2bc2a9: Pushed
de43e94a3438: Pushed
f3ba2250c524: Pushed
d915f35567d3: Pushed
be7084d609f6: Pushed
994b958a9456: Pushed
v1: digest: sha256:0f194559a650670e552404572dfeacec05296ac2750c395633afb15643bbae11 size: 856
```

```
(base) PS C:\Users\maazb\OneDrive\Desktop\Cloud-Dev\Assignments\Assignment4\CS487-PA4> docker push pa426100414.azurecr.io/report-job:v1
The push refers to repository [pa426100414.azurecr.io/report-job]
3d855105a4a0: Pushed
f3ba2250c524: Mounted from validate-api
91ff8760033c: Mounted from validate-api
7ccd73948dde: Mounted from validate-api
de43e94a3438: Mounted from validate-api
22d84cb0741a: Pushed
934d1e46c047: Pushed
3531af2bc2a9: Mounted from validate-api
93021b56d2d2: Pushed
v1: digest: sha256:27e4ad223c58771ae560878ff53686b20c53e7f4da2e62ff1e0a62d76862615f size: 856
```

```
(base) PS C:\Users\maazb\OneDrive\Desktop\Cloud-Dev\Assignments\Assignment4\CS487-PA4> docker push pa426100414.azurecr.io/func-app:v1
The push refers to repository [pa426100414.azurecr.io/func-app]
8b5c9ed400b7: Pushed
4f4fb700ef54: Pushed
af5ac245ac18: Pushed
48805a8c2422: Pushed
265a95973510: Pushed
295675d49bd5: Pushed
cc2ef16d9998: Pushed
2cfff7b2fc77f: Pushed
1e291ff5c47c: Pushed
6d411a239a49: Pushed
c2b164019d21: Pushed
83dc37193a82: Pushed
5999d6a7b922: Pushed
66f110fb4b1b: Pushed
8715c03abd5b: Pushed
f6e58f04b6b2: Pushed
40104569dda3: Pushed
60993470af6a: Pushed
3263c6727b31: Pushed
9e50f0fc8e7b: Pushed
3b3815156222: Pushed
89320e7119e2: Pushed
v1: digest: sha256:958dfe01a60bffc3624b8ed19c1f6b46adf5daadecdd379d5162e16be9349276 size: 856
```

Task 3:

Durable Functions visible on func start:

```
(.venv) (base) PS C:\Users\maazb\OneDrive\Desktop\Cloud-Dev\Assignments\Assignment4\CS487-PA4\function-app> func start

Azure Functions Core Tools
Core Tools Version:      4.10.0+2ced9a89e29b04615bc55cd527c074250266362f (64-bit)
Function Runtime Version: 4.1048.200.26180

[2026-05-06T07:39:45.132Z] Durable Functions Distributed Tracing V2 is GA now! Learn how to enable the feature by visiti
ng aka.ms/durable-distributed-tracing. To disable this message, you can configure distributedTracingEnabled to "true" an
d version to "V2" or "None". Setting it to "None" would in effect disable the feature.
[2026-05-06T07:39:45.432Z] Worker process started and initialized.

Functions:

    http_starter: [POST] http://localhost:7071/api/orchestrators/my_orchestrator

    my_orchestrator: orchestrationTrigger

    report_activity: activityTrigger

    validate_activity: activityTrigger

For detailed output, run func with --verbose flag.
```

Task 4:

Container Image Configuration:

"statusQueryGetUri":

"https://pa4-26100414-fn-a5ereucz9cue4cj.swedencentral-01.azurewebsites.net/runtime/webhooks/durabletask/instances/50a12e51df4e4bc795e268d67f5a8a43?taskHub=pa426100414fn&connection=Storage&code=YMHgkSKvnrXVBxVmF1KXzD5W6SPfHyansFxtT4evhLUUAzFucmpr9g=="

"Id":

"50a12e51df4e4bc795e268d67f5a8a43"

Status URL Query Json:

```
name: "my_orchestrator"
instanceId: "50a12e51df4e4bc795e268d67f5a8a43"
runtimeStatus: "Completed"
input: '{"order_id": "FINAL-SMOKE-TEST", "items": [{"sku": "ABC", "qty": 1}]}'
customStatus: null
output: null
createdTime: "2026-05-06T18:46:09Z"
lastUpdatedTime: "2026-05-06T18:46:10Z"
```

Task 5

Kubectl get nodes:

```
(base) PS C:\Users\maazb\OneDrive\Desktop\Cloud-Dev\Assignments\Assignment4\CS487-PA4> kubectl get nodes
NAME                                STATUS    ROLES    AGE    VERSION
aks-nodepool1-34692032-vmss000000  Ready    <none>   17m    v1.34.6
```

Kubectl get pods:

```
(base) PS C:\Users\maazb\OneDrive\Desktop\Cloud-Dev\Assignments\Assignment4\CS487-PA4> kubectl get pods
NAME                                READY    STATUS    RESTARTS    AGE
validate-deployment-577f889bd7-d65mj  1/1      Running   0            14m
```

Kubectl get service:

```
(base) PS C:\Users\maazb\OneDrive\Desktop\Cloud-Dev\Assignments\Assignment4\CS487-PA4> kubectl get service
NAME            TYPE        CLUSTER-IP    EXTERNAL-IP    PORT(S)          AGE
kubernetes      ClusterIP   10.0.0.1      <none>         443/TCP          20m
validate-service LoadBalancer 10.0.217.194  20.90.53.87    8080:32749/TCP   15m
```

Good order:

```
(base) PS C:\Users\maazb\OneDrive\Desktop\Cloud-Dev\Assignments\Assignment4\CS487-PA4> Invoke-RestMethod -Uri "http://20.90.53.87:8080/validate" `
>> -Method Post `
>> -ContentType "application/json" `
>> -Body '{"order_id": "O-1001", "items": [{"sku": "ABC", "qty": 2}]}'

valid reason order_id
-----
True ok      O-1001
```

Bad order:

```
(base) PS C:\Users\maazb\OneDrive\Desktop\Cloud-Dev\Assignments\Assignment4\CS487-PA4> Invoke-RestMethod -Uri "http://2090.53.87.8080/validate" `
>> -Method Post `
>> -ContentType "application/json" `
>> -Body '{"order_id":"0-1002","items":[{"sku":"ABC","qty":999}]}'

valid reason      order_id
-----
False quantity exceeds limit 0-1002
```

Function App Application Setting (Validate URL):

pa4-26100414-fn | Environment variables

App settings

Search

+ Add Refresh Show values Advanced edit Pull reference values

Name	Value	Deployment slot setting	Source	Delete
AzureWebJobsStorage_accountName	Show value		App Service	
AzureWebJobsStorage_clientId	Show value		App Service	
AzureWebJobsStorage_credential	Show value		App Service	
DOCKER_REGISTRY_SERVER_PASSWORD	Show value		App Service	
DOCKER_REGISTRY_SERVER_URL	Show value		App Service	
DOCKER_REGISTRY_SERVER_USERNAME	Show value		App Service	
FUNCTIONS_EXTENSION_VERSION	Show value		App Service	
VALIDATE_URL	Show value		App Service	
WEBSITES_ENABLE_APP_SERVICE_STORAGE	Show value		App Service	

Task 6

reports blob creation:

pa426100414 | Containers

Storage account

Search

+ Add container Upload Refresh Delete Change access level Restore containers

Search containers by prefix

Showing all 5 items

	Name	Last modified	Anonymous access level	Lease state
	\$logs	07/05/2026, 12:42:02 am	Private	Available
	azure-webjobs-hosts	06/05/2026, 9:54:24 pm	Private	Available
	azure-webjobs-secrets	06/05/2026, 9:54:17 pm	Private	Available
	pa426100414fn-applease	06/05/2026, 9:54:19 pm	Private	Leased
	reports	07/05/2026, 8:59:06 am	Private	Available

Successful container show (for ci-report-test):

```
maaz [ ~ ]$ az container show --resource-group rg-sp26-26100414 --name ci-report-test --query "instanceView.state" --output tsv
succeeded
```


Container logs terminal output:

```
maaz [ ~ ]$ az container logs --resource-group rg-sp26-26100414 --name ci-report-test
Uploaded TEST-001.pdf to reports container
```

PDF in reports container:

Home > Storage center > Blob Storage > pa426100414 > Containers

reports

Container

Search

+ Add Directory Upload Change access level Refresh Delete Copy Paste Rename Acquire lease Break lease Edit columns

Overview

Diagnose and solve problems

Access Control (IAM)

Settings

reports

Authentication method: Microsoft Entra user account (Switch to Access key)

Add filter

Search blobs by prefix (case-sensitive)

Only show active blobs

Showing all 1 items

Name	Last modified	Access tier	Blob type	Size	Lease state
TEST-001.pdf	07/05/2026, 9:08:38 am	Hot (inferred)	Block blob	1.43 KiB	Available

Function App Identity Blade (showing Managed Identity):

pa4-26100414-fn | Identity

Function App

Search

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Microsoft Defender for Cloud

Events (preview)

System assigned

User assigned

User assigned managed identities enable Azure resources to authenticate to cloud services (e.g. Azure Key Vault) without storing credentials in code. This type of managed identities are created as resources, and have their own lifecycle. A single resource (e.g. Virtual Machine) can utilize multiple user assigned managed identities. Similarly, a single user assigned managed identity can be shared across resources (e.g. Virtual Machine).

+ Add Remove Refresh Troubleshoot Got feedback?

Name	Resource group	Subscription
mi-pa4-26100414	rg-sp26-26100414	67e93b84-fe08-452c-80ea-175d0a3eca56

Function App settings

REPORT_*:

REPORT_IMAGE	Show value	App Service	
REPORT_LOCATION	Show value	App Service	
REPORT_RG	Show value	App Service	

ACR_*:

ACR_PASSWORD	Show value	App Service	
ACR_SERVER	Show value	App Service	
ACR_USERNAME	Show value	App Service	

STORAGE_ACCOUNT_URL:

STORAGE_ACCOUNT_URL	Show value	App Service	
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Task 7

Filled Form:

TaskFlow

Order ID

ORD-001

SKU

WIDGET-X

Quantity (Max 100)

2

Submit Order

Function Invocations (Log Stream):

```
maaz [ ~ ]$ az webapp log tail --name pa4-26100414-fn --resource-group rg-sp26-26100414
2026-05-07T15:15:26 Welcome, you are now connected to log-streaming service.
Starting Log Tail -n 10 of existing logs ----
/appsvctmp/volatile/logs/runtime/container.log
2026-05-07T14:54:36.0640247Z info: Function.my_orchestrator[2]
2026-05-07T14:54:36.0640506Z Executed 'Functions.my_orchestrator' (Succeeded, Id=432efd2e-a925-43fd-90f7-926b812b0b56, Duration=37ms)
2026-05-07T14:54:36.0646806Z MS_FUNCTION_AZURE_MONITOR_EVENT 4,pa4-26100414-fn-a5ereucz9c9e4cj.swedencentral-01.azurewebsites.net,Microsoft.Web/sites/functions/log,FunctionAppLogs,swedencentral,{'appName':'pa4-26100414-fn','roleInstance':'9d4c7e962a9486d3351a27c2a032aad96892fc125f45ab5254d12169dae7fd2f','message':'Executed Functions.my_orchestrator (Succeeded, Id=432efd2e-a925-43fd-90f7-926b812b0b56, Duration=37ms)','category':'Function.my_orchestrator','hostVersion':'4.1048.200.5','functionInvocationId':'432efd2e-a925-43fd-90f7-926b812b0b56','functionName':'Functions.my_orchestrator','hostInstanceId':'4c4d2f61-e59e-4aed-869c-1611d839274f','level':'Information','levelId':2,'processId':9,'eventId':2,'eventName':'FunctionCompleted'}",05/07/2026 14:54:36
2026-05-07T14:54:39.3156908Z info: Microsoft.Azure.WebJobs.Hosting.OptionsLoggingService[0]
2026-05-07T14:54:39.3157179Z ExtensionSystemOptions
2026-05-07T14:54:39.3157201Z {
2026-05-07T14:54:39.3157216Z   "extensionName": "durabletask",
2026-05-07T14:54:39.3157231Z   "webhookAuthorizationLevel": "System"
2026-05-07T14:54:39.3157277Z }
2026-05-07T14:54:39.3230173Z MS_FUNCTION_AZURE_MONITOR_EVENT 4,pa4-26100414-fn-a5ereucz9c9e4cj.swedencentral-01.azurewebsites.net,Microsoft.Web/sites/functions/log,FunctionAppLogs,swedencentral,{'appName':'pa4-26100414-fn','roleInstance':'9d4c7e962a9486d3351a27c2a032aad96892fc125f45ab5254d12169dae7fd2f','message':'ExtensionSystemOptions\n{\n  \"extensionName\": \"durabletask\",
  \"webhookAuthorizationLevel\": \"System\"\n}','category':'Microsoft.Azure.WebJobs.Hosting.OptionsLoggingService','hostVersion':'4.1048.200.5','hostInstanceId':'4c4d2f61-e59e-4aed-869c-1611d839274f','level':'Information','levelId':2,'processId':9},"05/07/2026 14:54:39
Ending Log Tail of existing logs ---
Starting Live Log Stream ---
2026-05-07T15:16:26 No new trace in the past 1 min(s).
```

